Deep Learning for Economics and Finance

Key takeaways, method choice, and outlook

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Closing lecture: climate economics, deep UQ, and policy design

Materials: https://github.com/sischei/Deep_Learning_for_Solving_And_Estimating_Dynamic_Economic_Models

Wrap-Up Roadmap

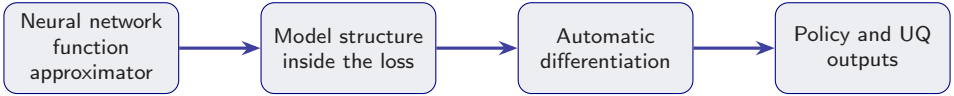
1. **Synthesis of the full course pipeline**
2. **Decision guide: DEQN, PINN, Surrogates, GP**
3. **Climate-policy lessons**
4. **Implementation checklist and outlook**

Learning objectives

By the end of the course you should be able to:

- Choose between DEQN, PINN, deep surrogates, and GP+BAL based on problem structure and scalability
- Design residual-loss functions that embed economic and climate equations for end-to-end equilibrium learning
- Articulate when to include parameters as pseudo-states versus solving repeatedly
- Evaluate whether a trained surrogate meets publication standards through residual diagnostics and stress tests
- Synthesize deep learning, uncertainty quantification, and constrained optimization for evidence-based policy design

The Big Picture: One Computational Paradigm



Across methods

DEQN, PINN, deep surrogates, and GP-BAL differ in objective and geometry, but share the same core optimization logic.

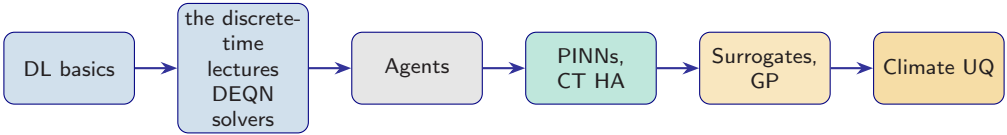
Course Arc

Day	Core step in the pipeline
1	Deep learning foundations and optimization diagnostics
2–4	DEQNs for discrete-time equilibrium (RA, IRBC, OLG, heterogeneity)
5	PINNs for continuous-time PDE problems
7	Deep surrogates and GP/BAL for fast policy/estimation loops
8	Climate IAMs, deep UQ, constrained policy design

Main message

The toolkit is modular: solve, emulate, quantify uncertainty, then optimize policy under explicit constraints.

Visual Timeline of the Course Workflow



Core paradigm:
model structure in loss + autodiff + scalable function approximation

Decision Guide: Which Method When?

Method	Use case	Strength
DEQN	Discrete-time equilibrium systems	Scales to high-dimensional states
PINN	Continuous-time PDE/HJB/KFE systems	Mesh-free PDE residual training
Deep surrogate	Repeated model evaluations	Very fast policy and estimation loops
GP + BAL	UQ and active design	Built-in uncertainty and sample efficiency

Climate UQ in One Slide

Climate UQ in context. Pseudo-states let one trained DEQN answer policy questions for *all* plausible parameter values simultaneously. A GP surrogate on top enables fast computation of SCC, temperature paths, and welfare across the parameter space. Bayesian active learning then identifies which parameters drive policy recommendations, for example, climate sensitivity matters more than the pure rate of time preference for carbon-tax design.

The payoff

Constrained Pareto-improving tax rules become feasible once the expensive solves are amortized.

Common Failure Modes and Mitigations

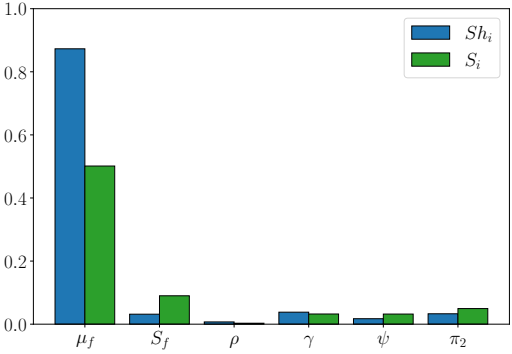
Failure modes

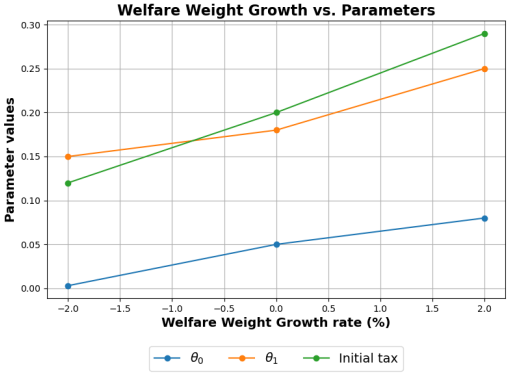
- Calibration mismatch to benchmark science
- Training focused on wrong state regions
- Over-reliance on mean loss diagnostics
- Surrogate extrapolation outside training support

Mitigations

- Explicit benchmark tests and stress scenarios
- Mixed exogenous/endogenous sampling
- Report residual distributions and constraints
- Active learning and uncertainty-aware evaluation

Representative Constraint and Uncertainty Results





Left: SCC uncertainty decomposition (Friedl et al., 2023). Right: constrained policy frontier (Kübler, Scheidegger, and Surbek, 2025, arXiv:2507.01704).

Implementation Checklist

1. Define state/control vectors and feasibility transforms.
2. Engineer residual blocks from economics and climate equations.
3. Set sampling scheme and training diagnostics before tuning architecture.
4. Validate on out-of-sample simulations and benchmark moments.
5. Build surrogate/UQ layer only after solver quality is verified.
6. Separate normative objective from political/participation constraints.

Research Frontier

- Richer heterogeneity: regional, sectoral, and household dimensions.
- Endogenous innovation and technology diffusion under climate policy.
- Multi-model and model-ensemble learning for robust policy design.
- Tight integration of Bayesian learning and policy commitment under ambiguity.
- Better theoretical guarantees for convergence and error bounds.

Reading and Notebook Map

Climate readings

- `lectures/day8/readings/CDICE_Restud_production.pdf`
- `lectures/day8/readings/CDICE_Restud_production_appendix.pdf`
- `lectures/day8/readings/DeepUQ_with_an_application_to_IAM.pdf`
- `lectures/day8/readings/JPE_Macro.pdf`

Climate notebooks (suggested order)

- `01_Climate_Exercise.ipynb` (NumPy warm-up: carbon cycle, damages, BAU vs mitigation)
- `02_DICE_DEQN_Library_Port.ipynb` (deterministic CDICE-DEQN, library port)
- `03_Stochastic_DICE_DEQN.ipynb` (AR(1) shocks, Gauss–Hermite, SCC fan charts)

Closing Message

From method to policy

Deep learning methods are most useful when they improve economic credibility, computational scalability, and policy transparency at the same time.

Thank you for participating in the course.

Questions and discussion